

KALPAMANI

AI Swing & Momentum Trading System - Blueprint V3.0

Self-Maturing, Deterministic Long/Short Trading Platform

PROPOSED - NOT YET REPOSITORY AUTHORITY

This document is the proposed successor to Blueprint V2.1. It becomes binding only after owner approval, a documentation-only adoption PR, and corresponding updates to CLAUDE.md and the authority order. It does not authorize broker, provider, strategy, Paper, live, or capital actions by itself.

V3 END-STATE VISION

An autonomous quantitative platform that discovers, ranks, researches, challenges, allocates, trades, manages, reconciles, measures, and improves. The system is self-maturing but not self-governing: it may diagnose, research, challenge, shadow, recommend, and fail closed automatically; humans authorize strategy releases, capital, leverage, licences, and live scaling.

Drafted from Blueprint V2.1, the Phase-3 planning package, the live Phase-1/2 evidence, the current Phase-3A A1 implementation, independent strategy review, and owner decisions recorded through August 27, 2026.

Personal-use system | USD 80,000 allocated strategy capital | QuantConnect LEAN | IBKR Pro | Long + Short

Version: Proposed 3.0 | Date: August 27, 2026

Document Control

Field	Value
Document	KalpaMani Blueprint V3.0
Status	PROPOSED - owner review required
Supersedes	Blueprint V2.1 only after formal repository adoption
Current repository main	9ffeea2f57de1dec233937e1627d2acb27fb1051
Current implementation PR	#7 - Phase 3A A1 foundation kernel - OPEN / UNMERGED
Current PR #7 head	6b680c03b85f21ce8d3703c1181c07a6e4bfa3bc
Current project phase	Phase 3A A1 - in review, not accepted
Strategy capital	USD 80,000, separate from broker-reported equity
Repository visibility	PUBLIC during development; PRIVATE required before micro-live/real money
Authority when adopted	Blueprint V3.0 -> approved ADRs -> CLAUDE.md -> task specs -> implementation judgment

Source Basis

- Blueprint V2.1 (updated August 24, 2026), used as the structural baseline and operational safety source.
- Merged Phase-3 point-in-time planning package and proposed ADR-0005.
- Accepted Phase-1 and Phase-2 certifications, including the empirical broker-side-control erratum and restart identity findings.
- Current Phase-3A A1 implementation and PR #7 review record.
- Independent strategy-engine review, used as a hypothesis and risk-analysis source rather than a binding specification.
- Owner decisions: personal use only; long and short; USD 1,500 annual base data budget; conditional expansion up to USD 5,000; QuantConnect retained as engine; Sharadar/EDGAR preferred data architecture; options chain as a future research/risk overlay; self-maturing but human-governed strategy evolution.

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1. V3 Change Summary

Blueprint V3.0 preserves the strongest safety and operational decisions of V2.1 while replacing the parts that our data, execution, licensing, and strategy discussions have matured beyond. V3 locks the architecture for evolution; it does not lock permanent strategy parameters or claim that any alpha has already been proven.

Area	V3 decision
System maturity	Self-maturing but not self-governing. Automatic diagnosis, research, challengers, shadowing, health actions, and fail-closed reduction; human approval for order-producing releases, capital, leverage, licences, and live scaling.
Strategy taxonomy	Separate alpha family, strategy module, trade template, feature, filter, and risk overlay. Different names do not create diversification.
Core strategy organization	Momentum Breakout and Momentum Pullback remain separately versioned and attributed strategy modules inside a shared Momentum Continuation family and factor-risk budget. PEAD remains a separate Event Drift family. A Fundamental Deterioration Short engine is added.
Shorts	Short alpha is structurally asymmetric: deterioration/event evidence + negative relative/residual momentum + breakdown/failed-rally timing + mandatory borrow/squeeze/SSR/recall controls.
Data	Vendor-neutral Bronze/Silver/Gold PIT platform with explicit public/provider/system availability profiles and immutable lineage.
Provider roles	QuantConnect LEAN remains the engine. Sharadar + SEC EDGAR is the preferred primary historical PIT stack, subject to qualification/licensing. QuantConnect local data is not imported into KalpaMani; cloud-contained experiments may be used where licensing permits.
Budget	USD 1,500/year base data budget; up to USD 5,000/year conditionally for qualified, incremental data. Personal-use only.
Options	Not a core dependency and no automated options trading. Historical daily option-chain features may be tested later as an event, volatility, squeeze, and risk overlay.
AI	Research and Challenger only. AI may generate hypotheses and recommendations but cannot approve money, risk, strategy promotion, or broker actions.
Strategy evolution	Immutable versions, Champion-Challenger, open-position version pinning, strategy registry, hypothesis registry, health states, promotion/rollback/retirement gates.
Allocator	Initial family/strategy risk caps. No fast performance-chasing allocator. Slow allocation changes only after substantial forward evidence.
Repository security	PUBLIC during development. PRIVATE before micro-live, real money, or production broker credentials. Vendor data and subscribed-data-derived artifacts remain outside Git.

2. Mission and Locked Principles

KalpaMani trades liquid U.S. common stocks long and short over a primary 2-30 trading-day horizon. It automatically discovers, ranks, researches, challenges, enters, protects, manages, pyramids, exits, reconciles, attributes, and learns within explicit deterministic limits.

LOCKED PRINCIPLE 1 - MONEY AND BROKER CONTROL

AI may improve information processing, research, hypothesis generation, and challenge analysis. Deterministic software controls eligibility, ranking gates, portfolio constraints, position sizing, borrow validation, order approval, execution, stops, pyramiding, reconciliation, circuit breakers, and the kill switch.

LOCKED PRINCIPLE 2 - SELF-MATURING, NOT SELF-GOVERNING

KalpaMani may observe, diagnose, hypothesize, research, simulate, shadow, recommend, reduce exposure, suspend unsafe new entries, and fail closed automatically. It may not promote a new or modified strategy into order-producing Paper/live operation, increase capital or risk, enable leverage, purchase a licence, or resume a safety suspension without explicit human authorization.

LOCKED PRINCIPLE 3 - POINT-IN-TIME REALITY

Missing required data is a refusal, not a zero, neutral, empty, latest, or silently degraded answer. Every result is profile-keyed, as-of bounded, reproducible, and tied to immutable source and derived lineage.

- Broker-reported account equity never overwrites the human-authorized USD 80,000 strategy capital.
- No averaging down. Pyramiding only into confirmed winners under deterministic rules.
- Initial leverage is none.
- Strategy versions, factor definitions, risk policies, model versions, prompt versions, code commits, and open-position ownership are immutable and auditable.
- No strategy is authorized because code exists. Authorization is a separate registry/governance state.
- A public repository may contain code and synthetic fixtures, never credentials, brokerage identifiers, vendor payloads, reconstructable subscribed data, or production configuration.

3. Current Project Baseline

Area	Current verified/governed state
Phase 1	Complete and accepted: LEAN -> IBKR Paper connectivity, market data, account-state observation, USD 80,000 capital separation, zero orders/positions.
Phase 2	Complete and accepted: one-share SPY Paper lifecycle, actual fill, protection, restart, broker-native identity recovery, controlled exit, flat reconciliation.
Phase 3 planning	Accepted and merged; ADR-0005 remains Proposed.
Phase 3A A1	Open PR #7; vendor-neutral, synthetic-only PIT foundation kernel; in review and not accepted.
Strategies	Not implemented and not authorized.
AI agents	Not implemented and not authorized.
Portfolio/risk engine	Not implemented beyond parameters and Phase-2 execution envelope.
Broker	IBKR Pro behind adapter; Paper only; live hard-disabled.
Engine	QuantConnect LEAN already selected, used, and certified with IBKR Paper.
Provider data	No real provider connected; no production data; G1-G5 remain open.
Repository	PUBLIC for development; must return PRIVATE before real money/micro-live.

V3 does not invalidate Phase 3A A1. The current kernel is the correct vendor-neutral foundation and should be finished, independently reviewed, accepted, and merged before the repository adopts this blueprint as authority.

4. End-to-End Architecture

DATA -> PIT PLATFORM -> UNIVERSE / FACTOR MATRIX
 v
 STRATEGY BRAIN: alpha -> module -> template -> consolidate
 -> AI Research -> AI Challenger -> deterministic compiler
 v
 CANDIDATE INTENT (NO SIZE, NO ORDER)
 v
 PORTFOLIO / FAMILY CAPS -> EVENT / GAP / BORROW -> RISK / SIZE
 v
 LEAN / BROKER -> FILL PROTECTION -> POSITION / RECONCILIATION
 v
 JOURNAL / ATTRIBUTION / HEALTH -> RESEARCH / GOVERNANCE

The architecture deliberately separates five questions: why an opportunity exists, why to enter now, whether the portfolio can own it, how much risk is allowed, and how the broker should execute it. No one module is allowed to answer all five.

5. Environments and Human Control Plane

Environment	Brokerage	Orders	Purpose
RESEARCH	None	None	PIT research, factors, strategy tests, shadow reconstruction.
SHADOW	None	None	Daily production-like candidates and simulated lifecycle; no broker.
PAPER	IBKR Paper	Only authorized strategy versions and run-scoped execution paths	Operational validation and automated Paper certification.
MICRO-LIVE	IBKR live	Tiny capital, no leverage, separately authorized	Prove real fills, slippage, routing, borrow, partial fills, and operations.
SCALED LIVE	IBKR live	Progressive capital only after review	Mature autonomous production.

5.1 Human authority

- Approve Blueprint/ADR changes, provider licences, strategy promotions, parameter releases, capital scaling, leverage, and live-risk changes.
- Maintain broker-required authentication/session operations and an independent kill switch.
- Review material safety exceptions and approve resumption after a safety suspension.
- Do not approve normal individual trades in mature production.

5.2 Automatic authority

- Trade approved strategy versions within deterministic limits.
- Reduce/disable new entries under approved health, data, risk, or reconciliation rules.
- Continue deterministic safe position management when AI is unavailable.
- Research, diagnose, generate challengers, shadow, and prepare governance recommendations.

6. Point-in-Time Data and Provider Architecture

6.1 Information-set profiles

Profile	Meaning	Primary use
PUBLIC_PIT	What was publicly available by the cutoff.	Sensitivity analysis and public-information studies.
PROVIDER_REALISTIC_PIT	What the selected provider could have	Primary historical capital-informing

Profile	Meaning	Primary use
	delivered by the cutoff, using exact timing or approved conservative bounds.	research, if provider semantics qualify.
FORWARD_SYSTEM	What KalpaMani actually received and when.	Forward Paper, monitoring, and live evidence.

No research accessor has a default profile, default as-of time, or latest/current shortcut. A global downgrade, per-dataset bound, or exclusion is explicit and enters run identity.

6.2 Storage layers

Layer	Contract
Bronze	Immutable content-addressed vendor bytes plus immutable acquisition records; no overwrite.
Silver	Internal security identity, UTC instants, explicit revisions, normalized units, provenance retained.
Gold	Profile-keyed curated facts and derived artifacts: raw bars, actions, universe snapshots, factor-ready datasets, reproducible adjusted artifacts.
Operational DB	PostgreSQL for signals, strategy registry, trades, audit state, health, approvals, and transactional operations.

6.3 Provider roles and budget

Domain	V3 role
Engine/backtests/executions	QuantConnect LEAN - already selected and used.
Broker	IBKR Pro behind BrokerAdapter.
Primary historical PIT foundation	Sharadar + SEC EDGAR is the preferred architecture, subject to qualification and personal-use licence verification at purchase.
QuantConnect local historical data	Not imported into KalpaMani Bronze/Silver/Gold. Local-use restrictions and cost make it unsuitable as the primary PIT store.
QuantConnect Cloud options research	Possible later cloud-contained A/B experiments; no raw export into KalpaMani unless rights permit.
Historical analyst revisions	Optional high-value data; purchase only if genuine PIT, adequate coverage, licence acceptable, and price justified.
Historical borrow	Required to certify short research. Qualify free/low-cost paths first; no short backtest claim without qualified evidence.
Budget decision	Locked V3 policy
Base annual data budget	USD 1,500
Conditional expanded ceiling	Up to USD 5,000/year only for a qualified dataset with measurable incremental value
Usage	Personal use and owner capital only
Spending priority	PIT foundation -> short borrow -> analyst revisions -> independent validation -> optional enhancements

DATA LICENSING RULE

Vendor payloads, normalized vendor-derived records, reconstructable derivatives, credentials, and vendor-quality reports stay under untracked runtime storage. Public code may describe interfaces and synthetic fixtures, not subscribed data.

7. Tradable Universe

Filter	Initial V3 research value
Security	NYSE/NASDAQ U.S. common stocks
Price	> USD 10
Market capitalization	> approximately USD 1.5B
Average daily dollar volume	> approximately USD 25M
History	> 250 trading days
Initial exclusions	OTC, warrants, preferreds, leveraged ETFs, microcaps, extremely illiquid binary names, and extreme HTB/squeeze

Filter	Initial V3 research value
	candidates

These values are versioned research starting points, not proven constants. Historical universe membership is materialized per session, profile, and definition version and is never reconstructed from the current universe.

8. Strategy Brain Architecture

```

POINT-IN-TIME REALITY GATE
  v
DETERMINISTIC FACTOR MATRIX
  v
ALPHA FAMILY + STRATEGY MODULES
  v
ENTRY-TEMPLATE MATCHING
  v
CANDIDATE CONSOLIDATION / FACTOR OVERLAP
  v
AI RESEARCH -> AI CHALLENGER
  v
DETERMINISTIC DECISION COMPILER
  v
READY_FOR_RISK_REVIEW / WATCHLIST / REJECT / BLOCKED

```

8.1 Reality gate

- Proves the security, universe, ticker, sessions, bars, events, filings, corporate actions, revisions, and required quality evidence were admissible at the cutoff.
- Refuses missing required inputs; optional omissions are explicit and versioned.
- Carries resolved profile, dataset versions, quality report, and complete lineage into every candidate.

8.2 Shared deterministic factor matrix

Factor family	Candidate measures
Price/relative/residual momentum	20/60/126-day total return, SPY-relative, sector-relative, market/sector residual momentum, trend persistence, 52-week-high proximity.
Earnings/event	SUE, revenue surprise, guidance direction, abnormal return/CAR, abnormal volume, margin acceleration, later genuine PIT revisions.
Price/volume quality	Base compactness, volatility contraction, pullback quality, relative volume, trend stability, liquidity, gap behavior.
Fundamental quality/deterioration	Sales/margin/cash-flow direction, accruals, leverage/distress, share-count changes, profitability trend.
Risk/context	Market regime, sector/correlation, gap risk, borrow/squeeze, event schedule, later options volatility/skew.

Residualization is an A/B research candidate, not a doctrine. V3 tests raw momentum, relative momentum, and market/sector residual momentum and retains whichever survives out-of-sample evidence and capacity tests.

8.3 Candidate consolidation

A security qualifying through residual momentum, industry strength, 52-week-high proximity, and a breakout is one economic opportunity with several supporting features - not four independent trades. The Brain consolidates duplicate economic exposure before portfolio risk is considered.

8.4 AI Research and Challenger

Agent	Allowed role	Forbidden role
Research Agent	Extract structured facts from approved candidate documents; assess catalyst,	Universe scanning, position size, risk override, arbitrary trade generation,

Agent	Allowed role	Forbidden role
	demand, margins, competitive position, management tone, and risks with source/timestamp provenance.	broker action.
Challenger Agent	Attempt to falsify the thesis; surface contradiction, crowding, valuation, event, dilution, alternative explanations, and evidence quality.	Approve trades or weaken deterministic rules.

A deterministic candidate that failed cannot be rescued by AI. AI influence is bounded, versioned, and measured through A/B/C testing: deterministic only; structured evidence; structured evidence plus LLM interpretation.

8.5 Decision compiler

- Validates data and provenance, authorized strategy/model versions, entry setup, AI schemas, unresolved contradictions, market permission, and duplicate economic exposure.
- Returns a typed status: READY_FOR_RISK_REVIEW, WATCHLIST, REJECTED, BLOCKED_DATA, BLOCKED_EVENT, BLOCKED_AI, BLOCKED_CONTRADICTION, or BLOCKED_BORROW.
- Never returns shares, dollars, broker route, or final order instructions.

9. V3 Strategy Taxonomy and Core Engines

9.1 Taxonomy

Type	Meaning	Examples
Alpha family	Economic source of expected outperformance/underperformance.	Momentum Continuation; Event/Information Drift; Fundamental Deterioration.
Strategy module/book	Separately versioned, researched, attributed opportunity process.	Breakout Long; Pullback Long; PEAD Long; PEAD Short; Deterioration Short.
Trade template	Why enter now.	Volatility-contraction breakout; controlled pullback reversal; post-gap stabilization; failed rally; support breakdown.
Feature	Ranking information, not a separate capital allocation by default.	Residual momentum; industry strength; 52-week-high proximity; margin trend.
Filter	Eligibility decision.	Liquidity; earnings buffer; borrow availability; common-stock status.
Risk overlay	Risk permission or treatment, not primary alpha.	Market regime; gap risk; options IV/skew; correlation; squeeze risk.

9.2 Core Family A - Momentum Continuation

Momentum Breakout and Momentum Pullback remain separate, independently versioned and attributed strategy modules, but V3 places them inside one Momentum Continuation family with shared factor-exposure accounting and a common family risk cap. This preserves their distinct entries and trade economics without pretending their momentum exposure is independent.

Module	Core mechanism	Research focus
Breakout Long	Strong relative/residual momentum + compact base/resistance + price/volume confirmation.	False breakouts, slippage, gap risk, sector/factor crowding, entry timing.
Pullback Long	Established trend + controlled retracement + contracting volume + renewed demand.	Trend quality, reversal confirmation, stop efficiency, regime sensitivity.

A future breakdown-short template is not automatically authorized by inverting Breakout Long. Short use requires the V3 short architecture below.

9.3 Core Family B - Event and Information Drift

Module	Core mechanism	Status
PEAD Long	Positive earnings/revenue/guidance information + abnormal return/volume + post-gap stabilization/continuation.	Core research module.
PEAD Short	Negative surprise/guidance + negative abnormal return + failed stabilization + qualified borrow/squeeze risk.	Core research module, short controls mandatory.
Post-Guidance Drift	Management guidance changes and subsequent information diffusion.	PEAD subtype initially; independent only if trade population/economics justify.

9.4 Core Family C - Fundamental Deterioration Short

V3 adds a dedicated short-only engine. A short candidate requires negative fundamental/event evidence, negative relative/residual momentum, an execution trigger, and safe borrow/squeeze conditions. Bottom-decile momentum alone is not authorization.

- Negative earnings or guidance; margin/revenue/cash-flow deterioration; high accruals or dilution evidence.
- Negative relative/residual momentum and weak sector context.
- Breakdown, failed gap-up, or failed rally timing.
- Borrow availability, fee, shortable quantity, utilization/crowding, SSR, recall/buy-in, and binary-event controls.

9.5 Later candidates

Candidate	Initial classification
Peer Catalyst Momentum	Later strategy after deterministic peer mapping and AI evidence layer qualify.
Corporate Dilution / ATM Offering Drift	Research backlog; likely capital-structure-event short family or Deterioration subtype.
Residual Reversal	Potential diversifier; cost-sensitive and research-only.
Statistical Arbitrage / Pairs	Deferred; not needed for V3 core swing system.
Strategy performance allocator	Deferred until substantial forward evidence; no fast performance chasing.

10. Options-Chain Research and Risk Overlay

V3 OPTIONS DECISION

Options-chain information is a future research candidate and optional risk/event/squeeze overlay. It is not required by Phase 3, is not a core initial alpha engine, and does not authorize automated options trading.

Priority	Features
Tier 1 - test first	ATM IV; implied move; IV percentile/change; 25-delta put-call skew; 30/60-day term structure; changes in skew/term structure; pre-earnings implied move versus actual move; options liquidity.
Tier 2 - exploratory	Open-interest changes/concentration; abnormal option volume; tail-skew curvature; expected versus realized volatility.
Defer/reject for initial use	Dealer GEX assumptions; Max Pain; raw put/call volume as direction; simplistic call-OI bullish/put-volume bearish stories.

QuantConnect Cloud may support historical daily option-chain A/B experiments inside LEAN. Raw QC options data is not assumed exportable into KalpaMani. If the overlay proves useful, live option context may later be requested only for shortlisted candidates through an approved feed, not streamed across the entire universe.

Options features initially produce context states such as NORMAL, ELEVATED_EVENT_RISK, EXTREME_TAIL_PRICING, SQUEEZE_RISK, and OPTIONS_DATA_UNRELIABLE. Hard thresholds such as IVR > 80 are research hypotheses, not locked gates.

11. Portfolio, Risk, Gap, and Capital Allocation

11.1 Strategy capital

The authoritative strategy capital is USD 80,000. Broker account equity is observed for reconciliation only and cannot participate in sizing.

Control	Initial V3 research value
Long planned risk/trade	0.50% of allocated capital
Short planned risk/trade	0.25%
Maximum open planned risk	Approximately 5%
Maximum individual position	Approximately 8-10%
Initial gross short exposure	$\leq 25\%$
Portfolio volatility target	Approximately 10-12% annualized
Initial leverage	None
Averaging down	Disabled
Pyramiding	Confirmed winners only, with stop/open-risk reduction

11.2 Position sizing

```

risk_dollars = strategy_capital * allowed_risk
risk_per_share = abs(entry_reference - technical_invalidation)
base_shares = risk_dollars / risk_per_share

```

Then cap by:

- maximum position
- family/strategy risk budget
- sector and correlation exposure
- portfolio volatility
- liquidity and capacity
- event/gap budget
- borrow/short constraints

11.3 Gap/event risk

- Technical stop distance is not assumed to equal total loss risk.
- Every candidate carries event flags, overnight gap estimate, earnings-carry permission, and correlated gap cluster.
- Breakout/Pullback do not normally carry through earnings unless a separately validated rule permits it.
- PEAD enters after the information event, not before it.

11.4 Family and strategy caps

V3 starts with caps, not target weights. Breakout and Pullback share a Momentum Continuation family cap in addition to their own module caps. PEAD and Deterioration Short have independent caps. Unused risk remains cash.

A slow allocator may later use trailing expectancy, drawdown, opportunity count, volatility, and cross-strategy correlation only after minimum evidence thresholds. It cannot chase recent performance or alter capital beyond approved bounds.

12. Short-Side Architecture

```

BORROW / SQUEEZE ELIGIBILITY -> DETERIORATION / EVENT EVIDENCE
-> NEGATIVE RELATIVE / RESIDUAL MOMENTUM -> BREAKDOWN / FAILED RALLY
-> PORTFOLIO / GAP PERMISSION -> REDUCED SIZE -> PRE-SUBMIT BORROW RECHECK
-> ORDER / RECALL / BUY-IN / COVER LIFECYCLE

```

- Borrow is checked at signal time and immediately before submission.

- Borrow fee, shortable quantity, fee deterioration, utilization/crowding, SSR, recall/buy-in, corporate actions, and binary events are explicit states.
- Hard-to-borrow and squeeze-prone names are excluded initially.
- Historical short results are not certified unless borrow cost and availability are modeled conservatively from qualified data.
- Borrow/utilization thresholds are versioned research parameters, not architecture constants.

13. Execution and Autonomous Operations

Requirement	V3 rule
Order identity	Deterministic intent/execution/client IDs; durable write-ahead; broker-native identity capture; duplicate prevention.
Partial fills	Protect filled quantity immediately; never protect nonexistent quantity.
Restart/reconnect	Reconcile broker truth before progression; adopt only positively identified owned orders; ambiguity halts.
Cancellation/exit	Cancel and confirm protection before close; no account-wide Liquidate/CancelAll.
Reconciliation	Positions, cash, open orders, ownership, fills, and strategy version.
Failures	Broker/data/database/AI failures fail safely. Preserve deterministic exits where safe.
Authentication	Human operational responsibility; second execution username/session before live automation.
Independent kill switch	Available even when AI and normal application control are unavailable.

14. Strategy Lifecycle and Self-Maturation

14.1 Lifecycle

IDEA
 -> REGISTERED HYPOTHESIS
 -> TAXONOMY / OVERLAP REVIEW
 -> DATA FEASIBILITY
 -> BASELINE RESEARCH
 -> LOCKED OUT-OF-SAMPLE VALIDATION
 -> SHADOW
 -> AUTOMATED PAPER
 -> MICRO-LIVE CANARY
 -> SCALED / WATCH / SUSPENDED / RETIRED

A code module, backtest result, or AI recommendation does not change lifecycle state. Every transition requires immutable evidence and the appropriate authorization.

14.2 Immutable versioning and Champion-Challenger

- Production versions are immutable. A refinement creates a new Challenger version.
- Open positions remain governed by the exact strategy, factor, risk, entry, exit, prompt, and execution versions that opened them.
- Champion and Challenger run on identical opportunities for overlap, economics, factor exposure, slippage, and failure-regime comparison.
- Challengers may automatically progress through preapproved research and shadow stages, but not into order-producing Paper/live without human approval.

14.3 Self-maturation authority

Activity	Automatic	Human approval
Monitor health and drift	Yes	Review material exceptions

Activity	Automatic	Human approval
Detect factor overlap/failure clusters	Yes	Approve taxonomy/capital changes
Generate ideas and preregister experiments	Yes within approved research boundaries	Approve high-risk/new-data experiments
Run backtests, OOS, stress, and shadow challengers	Yes	Approve promotion beyond shadow
Reduce/disable new entries after safety or health triggers	Yes	Approve resume
Change production parameters/models	No	Required
Promote to Paper/micro-live/scaled	No	Required
Increase capital/risk/leverage/short exposure	No	Required
Purchase licence or add provider	No	Required

15. Strategy Health and Learning

Health state	Meaning
HEALTHY	Within expected economic and operational envelope.
WATCH	Material drift or limited evidence; no automatic risk expansion.
DEGRADED	Persistent performance, data, or execution deterioration.
NEW_ENTRIES_REDUCED	Approved deterministic reduction applied.
NEW_ENTRIES_DISABLED	No new risk; existing positions managed safely.
SUSPENDED	Safety/governance review required before resume.
RETIRED	No future entries; evidence retained permanently.

KalpaMani measures expectancy, drawdown, tail losses, turnover, slippage, opportunity count, capacity, factor exposure, cross-strategy correlation, regime behavior, borrow failures, data-quality incidents, AI contribution, and modeled-versus-realized execution. Degradation creates a research queue, not automatic parameter mutation.

TRADE / SIGNAL JOURNAL
 v
 ATTRIBUTION BY FAMILY / MODULE / TEMPLATE / FACTOR / AI / REGIME
 v
 HEALTH + DRIFT DETECTION
 v
 RESEARCH HYPOTHESIS / CHALLENGER
 v
 OOS -> SHADOW -> GOVERNANCE PACKET
 v
 HUMAN-AUTHORIZED RELEASE

16. Research and Validation Methodology

- PIT, survivorship-aware, revision-aware datasets and immutable research manifests.
- Simple baseline first; complexity must beat a named baseline.
- Walk-forward and locked out-of-sample periods.
- Purged/combinatorial validation where overlapping horizons require it.
- Parameter-neighborhood stability; no sharp optimum cliffs.
- Deflated performance metrics and recorded trial counts to control multiple testing.
- Commissions, spread, slippage, realistic fills, liquidity, market impact, borrow cost, and gap stress.
- Regime, sector, factor, capacity, and failure-mode decomposition.
- AI A/B/C incremental-alpha test; retain AI only if it improves out-of-sample economics or risk control.

Initial required experiments	Purpose
Raw momentum vs relative vs market/sector residual momentum	Determine whether neutralization improves crash behavior and economics without removing useful exposure.
Breakout vs Pullback vs immediate ranked entry	Measure whether entry modules create distinct trade populations and P&L.

Initial required experiments	Purpose
PEAD SUE only vs SUE + CAR + volume + guidance	Measure incremental event information.
Short engine with/without borrow/squeeze filters	Quantify whether operational short controls materially improve net results and tails.
Core strategy vs core + options overlay	Measure risk reduction, missed opportunity, and true incremental value.

17. Deployment Gates and Operational Maturity

Mode	Behavior	Exit gate
Research	No orders.	Strategies survive realistic PIT OOS tests and costs.
Shadow	Daily production-like candidates; no broker.	Stable candidates, data, manifests, attribution, and health.
Automated Paper	No per-trade human approval for authorized versions.	No duplicate/missing orders; partial-fill, restart, reconciliation, risk, and strategy behavior stable.
Micro-live	Small capital; no leverage; limited strategy set.	Real fills, slippage, borrow, routing, risk, health, and operations track expectations.
Scaled autonomous	Progressive capital.	Repeated statistical and operational review.
Controlled leverage	Volatility-targeted incremental leverage only.	Long live record, controlled drawdown, reliable operations, separate approval.

REPOSITORY VISIBILITY GATE

The repository must return to PRIVATE before micro-live, real-money trading, or production broker credentials/configuration. Current PUBLIC development status does not satisfy a live deployment gate.

18. Tool Stack and Budget Governance

Layer	V3 choice
Research/backtest/execution engine	QuantConnect LEAN
Broker	Interactive Brokers Pro behind BrokerAdapter
PIT data platform	KalpaMani Bronze/Silver/Gold; Parquet + DuckDB planned after A1/vendor-volume evidence; PostgreSQL for operational state
Primary historical data	Preferred Sharadar + SEC EDGAR, subject to qualification/licence
Live market data	Independent audited feed/IBKR as approved; no broker-data dependency for historical ranking
Borrow data	IBKR live plus qualified historical provider
Historical options research	QuantConnect Cloud-contained experiment, licensing and timing qualified
AI	Structured LLM Research + Challenger only
Dashboard	Health, exposures, positions, controls, research queue, governance packets, kill switch
Source control	GitHub; public development, private before live

The data budget is a ceiling, not a target. No purchase occurs until the dataset passes licence, point-in-time, coverage, quality, and incremental-value gates.

19. V3 Implementation Roadmap

Repository phase	Deliverable	V3 capabilities introduced
Current - Phase 3A A1	Vendor-neutral PIT kernel	Contracts, profiles, provenance, immutable storage, quality, manifests, synthetic proofs. No Brain/strategies.
After A1 acceptance	V3 adoption PR + ADR-0006	Make Blueprint V3 authoritative; lock Brain/self-maturation governance. Documentation only.
Remaining Phase 3A	Real security master, calendars, prices,	Foundation for momentum,

Repository phase	Deliverable	V3 capabilities introduced
	actions, universe	Breakout/Pullback, liquidity, and identity.
Phase 3B	EDGAR, fundamentals, earnings, guidance, optional estimates	Foundation for PEAD, guidance drift, deterioration features.
Phase 3C	Borrow qualification	Required short eligibility and research certification.
Phase 3D	LEAN export/integration and production data gates	Reproducible research/engine bridge.
Phase 4A	Brain contracts, registry, taxonomy, factor/overlap framework	CandidateIntent, StrategySpec, lifecycle, health models; no orders.
Phase 4B	Factor matrix and core strategy research	Momentum, Breakout, Pullback, PEAD baselines and OOS evidence.
Phase 4C	Short engine, guidance, options experiments	Deterioration Short, PEAD Short, optional options risk overlay.
Phase 5	Portfolio/risk/AI/self-maturation integration	Decision compiler, AI agents, Champion-Challenger, automated research/health.
Phase 6	Full-system automated Paper	Multiple strategies, positions, partial fills, stops, operations, dashboard, shadow challengers.
Phase 7	Micro-live and scaling	Human-approved real-money canary, progressive scaling, later controlled leverage.

The earliest Brain documentation may begin after PR #7 is accepted and merged. Brain code begins only after final Phase-3 data acceptance, because strategy health and PIT research are meaningless without trustworthy inputs.

20. Locked V3 Specification

Decision	Locked V3
Mission	Autonomous personal long/short U.S. equity swing/momentum system, 2-30 trading days.
Strategy capital	USD 80,000 authoritative allocation; broker equity informational only.
Initial leverage	None.
Core families	Momentum Continuation; Event/Information Drift; Fundamental Deterioration Short.
Core strategy modules	Breakout Long; Pullback Long; PEAD Long; PEAD Short; Deterioration Short.
Shared momentum exposure	Breakout/Pullback share factor accounting and family cap; separate attribution retained.
Scanner/factors	Deterministic Python/SQL/PIT factor matrix.
AI	Research + Challenger; bounded evidence with source/timestamp/model/prompt provenance.
Brain output	CandidateIntent only; no size or broker instruction.
Risk	Deterministic, including gap/event, factor/family, sector/correlation, borrow/squeeze, and circuit breakers.
Execution	QuantConnect LEAN -> BrokerAdapter -> IBKR Pro; deterministic IDs, protection, reconciliation.
Shorts	Asymmetric, deterioration/event conditioned, qualified borrow mandatory.
Options	Future optional research/risk overlay; no core dependency; no automated options trading.
Strategy evolution	Self-maturing but human-governed; immutable versions; Champion-Challenger; no automatic Paper/live promotion.
Capital allocation	Caps/opportunity usage first; slow allocator later after forward evidence.
Data	PIT Bronze/Silver/Gold, profile-keyed, immutable lineage, no silent missing-data substitution.
Provider split	QuantConnect engine; preferred Sharadar + EDGAR historical

Decision	Locked V3
	stack; borrow/revisions remain qualified decisions.
Budget	USD 1,500 base; up to USD 5,000 conditional; personal use only.
Repository	Public development; private before real money.
Human role	Govern models, releases, capital, licences, exceptions, authentication, resumption, and kill switch.

21. Open Decision Gates

Gate	Question	Required before
G1 Provider selection	Does the preferred historical stack pass P1-P9 provider tests?	Credentialed production ingestion and Phase-3 acceptance.
G2 Production information profile	Which profile governs capital-informing historical research by dataset?	Capital-informing backtests.
G3 Licensing	Do personal-use terms permit intended own-account use, retained artifacts, and public code?	Purchase/credential/fetch.
G4 Analyst revisions	Can genuine PIT revisions be obtained at acceptable cost/licence?	Revision-enhanced PEAD/deterioration strategies.
G5 Borrow history	Can historical availability/fees be qualified with adequate coverage and PIT behavior?	Certified short backtests and short strategy promotion.
G6 Options overlay	Does daily options information improve OOS risk/alpha after timing and licensing controls?	Any production influence from options data.
G7 Strategy taxonomy evidence	Do Breakout/Pullback produce sufficiently distinct economics to justify separate module budgets within the family?	Later allocator and capital-budget decisions.

22. Adoption and Migration Plan

1. Finish the current PR #7 review at its latest head. Do not add V3 Brain/strategy code to PR #7.
2. After PR #7 is accepted and merged, create a documentation-only branch and PR containing this Blueprint V3.0, ADR-0006 (Adopt Blueprint V3.0 and Strategy Brain/Self-Maturation Governance), and updates to CLAUDE.md authority order and current status.
3. Keep ADR-0005 Proposed until Phase-3 provider/profile/licensing gates are resolved. V3 adoption does not silently accept ADR-0005.
4. Record explicit V2.1-to-V3 supersession and preserve the V2.1 PDF as historical architecture evidence.
5. Continue Phase 3A/3B/3C/3D under V3 data and governance contracts.
6. Begin Phase-4 Brain implementation only after Phase-3 final acceptance and a separately authorized implementation specification.

ADOPTION EFFECT

Once accepted and merged, Blueprint V3.0 becomes the highest project authority. Existing accepted Phase-1/2 evidence remains valid. Empirical errata and approved ADRs continue to govern where they refine or correct operational assumptions.

Appendix A - CandidateIntent Contract

Field group	Required contents
Identity	candidate_id, security_id, as_of_time, direction, environment
Strategy	alpha_family, strategy_module, trade_template, strategy_version, factor_definition_version
Evidence	factor_vector, rank, setup quality, data coverage, source/lineage IDs, reason codes
AI	research output, challenger output, source publish times, model version, prompt version

Field group	Required contents
Trade thesis	entry condition, invalidation condition, technical stop reference, expected holding period
Risk context	upcoming event flags, gap-risk estimate, earnings-carry permission, liquidity, family/factor exposures, sector/correlation cluster
Short context	borrow required, borrow evidence, squeeze/SSR/recall/event constraints
Decision	typed status and deterministic decision reasons

Explicitly excluded: shares, dollar size, final order type, broker route, client/broker order ID.

Appendix B - StrategySpec Contract

Category	Required contents
Identity/status	strategy_id, family, version, lifecycle status, authorized environments
Direction/horizon	LONG/SHORT/BOTH, expected holding horizon
Data	required/optional domains, profile, revision view, coverage contracts
Logic	factor dependencies, trade templates, entry/invalidation/exit policies
Permissions	market/event/gap/borrow prerequisites
Risk tags	family/factor/sector exposures, expected capacity, risk-policy compatibility
Reproducibility	manifest version, code commit, configuration, model/prompt versions
Research governance	hypothesis, baseline, trial budget, success/failure criteria, promotion history

Appendix C - Automatic vs Human Authority Matrix

Decision/action	System may do automatically	Human approval required
Normal trades under approved strategy	Yes	No per-trade approval
Data/quality/health fail-closed block	Yes	Approval to resume if suspended
Research hypothesis generation	Yes	Only for costly/new-data/high-risk experiment registration
Backtest/OOS/stress/shadow	Yes within approved scope	Approve scope expansion
Research -> order-producing Paper	No	Yes
Paper -> micro-live	No	Yes
Production model replacement	No	Yes
Capital/risk/leverage increase	No	Yes
Provider licence/purchase	No	Yes
Emergency reduce/disable new risk	Yes	No; review afterward
Emergency flatten	Only under preapproved independent kill-switch design	Human always retains direct control

Appendix D - V2.1 to V3 Delta

V2.1	V3 evolution
Breakout, Pullback, PEAD as independent engines	Retained as separately versioned modules, but Breakout/Pullback grouped under Momentum Continuation for factor exposure and family risk.
Short breakdown as stricter mirror-like logic	Dedicated Fundamental Deterioration Short family; breakdown is timing, not sufficient alpha.
Strategy health mentioned operationally	Full health state machine, immutable versions, Champion-Challenger, research queue, promotion/rollback/retirement.
Autonomous system learns	Explicit self-maturing/not-self-governing authority matrix.
LEAN/QuantConnect data initially	QuantConnect locked as engine; vendor-neutral PIT platform;

V2.1	V3 evolution
	preferred Sharadar + EDGAR, subject to open gates.
Options positioning not V1	Still not core V1; defined future daily options risk/event experiment with constrained feature set.
Per-strategy risk caps then allocator	Adds family exposure caps and factor-overlap control; slow allocator only after forward evidence.
AI Research + Challenger	Retained, with added hypothesis generation/research automation but no automatic release authority.
Generic build roadmap	Mapped to current repository Phase 3A-D, Phase 4 Brain/strategies, Phase 5 risk/AI/self-maturation, Phase 6 Paper, Phase 7 micro-live.

FINAL V3 DESIGN OBJECTIVE

Build a quantitative trading platform that can improve its data, factors, strategies, execution, and operations through automated scientific work while remaining deterministic, point-in-time, auditable, and human-governed wherever money, risk, licences, and live authority are concerned.